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Global offshore wind energy resources using the new ERA-5 reanalysis

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Abstract

A global quantitative characterization of offshore wind power density is presented over the economic exclusive zones (EEZs), at annual and seasonal scales, based on the recently released ECMWF ERA-5 reanalysis. Compared to its predecessors, ERA-5 features increased spatial and temporal resolutions, along with improved model parameterizations and data assimilation methodologies. Consequently, the present study represents an updated benchmark of wind resources over the global oceans. Further robustness to this argument is provided by our results showing that of the EEZs with large wind energy potential are characterized by regional-scale circulation features (coastal jets and breezes, land-sea and topographic circulations), known require high model resolution to be accurately simulated. Additionally, it is demonstrated that larger errors (up to 50%) can result from estimating wind power density from logarithmically extrapolated near-surface wind speed when compared to using high-resolution model level information. Coarser (daily) temporal resolution is also found to cause widespread increases in offshore wind power density errors between 10% and 30% compared to higher (hourly) resolution data. Further leveraging on the high vertical resolution of ERA-5, we show the large benefit of future increases in hub-heights from 100 m to 250 m for global offshore wind resources. Such technological advance results in widespread gains (ranging between +5% and +50%) throughout the global EEZs and no losses for all seasons. Moreover, the gains are >+10% for all regions with large wind energy potential at 100 m and >+40% over the Arctic Pacific sector and in regions characterized by the presence of coastal low-level jets.

1. Introduction

In the last years, the offshore wind energy sector has been going through a fast and broad development which includes: funded and floating harvesting technologies (GWEC 2019, IEA 2019), the installed and projected capacities in wider regions, and the observational and modelling characterization of available wind resource (Keivanpour *et al* 2017, Soares *et al* 2017, 2019a, Bosch *et al* 2018, Lima *et al* 2020a). The improvements on offshore wind technology allowed a growth of the global offshore wind market close to 30% per year over the last decade (IEA 2019). In 2018, the global installed offshore wind power capacity was of 23GW in contrast with 3GW installed in 2010. More than 80% of the installed capacity is in Europe: nearly 8GW in United Kingdom, 6.5GW in

Germany and 3.6GW distributed between Denmark, Netherlands and Belgium.

The offshore funded turbines have hub-heights between 70 m and 120 m encompassing a wind power capacity between 3 MW and 8 MW. Regarding floating technologies, there are wind turbines with hub-heights in the range of 70 m to 100 m and wind power capacities between 3 MW to 6 MW. Currently, new turbines are under development and it is expected that hub-heights will soon reach around 150 m with a power capacity of ~12 MW (Wiser *et al* 2011, OECD/IEA 2013). Future technological advances point to larger wind turbines in size, allowing to attain wind power capacities between 15 MW and 20 MW. This will be possible based on the adaptation of floating foundation technologies used in the oil and gas sectors to the wind offshore industry.

The main advantages of offshore wind harvesting, when compared to onshore wind resources, are the lower variability and larger magnitude of offshore wind at hub heights, the possibility of building larger power plants and the reduction of the noise and visual impacts nearby population settlements (Kaldellis and Kapsali 2013).

Most of the offshore installed power capacity corresponds to funded towers and hence deployed in shallow waters, up to 60 m depth (Bosch *et al* 2018) and close to the shore. Beyond 60 m depths the floating turbines are the only feasible option, which may enable a vast expansion of the potential areas of wind harvesting. However, there are obvious spatial constraints to the deployment of wind floating towers such as: legislative limits to build windfarms beyond some kilometres off the shore (e.g. in the Netherlands and UK the limit is 22 km (DECC 2009, Loeff 2014), areas of intense ship movements, maritime protected areas. Additionally, offshore windfarms may only be deployed within the economic exclusive zone (EEZ) of each country, which corresponds to the area within a maximum of 370 km off the coast of each country's shoreline. Finally, it is accepted that for the current status of the offshore technologies wind harvesting infrastructure may only be placed at water depths up to 1 km (Bosch *et al* 2018, Dupont *et al* 2018). Despite the limitations, the floating technology clearly paves the way for a major increase of wind installation capacity. In fact, it is projected that the growth in offshore technology could supply more than the current global electricity production (IEA 2019).

A different type of limitation for the development of the wind energy sector is the accurate quantitative portrayal and forecast of wind fields. This is particularly challenging at typical hub heights where only scarce localized observations exist, and even more so for offshore locations. Wind estimates derived from remote sensing measurements are available over the global ocean. But global-scale satellite-based wind estimates are only obtainable at 10 m height (Atlas *et al* 2011). Consequently, vertical extrapolation to hub heights is required, which introduces additional uncertainties, especially in areas of stronger winds and large wind shear (Soares *et al* 2014, 2019b, Lima *et al* 2018, 2019). Furthermore, the available remote sensing wind products display known shortcomings in coastal areas (Tang *et al* 2004). Global reanalyses constitute an alternative source of wind information, including estimates at different heights. Reanalyses combine state-of-the-art numerical weather prediction models and observational datasets by using data assimilation techniques (Saha *et al* 2010, Dee *et al* 2011, Kobayashi *et al* 2015), providing a physically based depiction of the entire atmosphere, which is constrained and, to a large extent, synchronized with the real weather.

In recent years a number of studies have characterized the global offshore wind resource using different wind data and methodologies. Zheng *et al* (2018) rezoned the global offshore wind energy resources, using the ERA-Interim reanalysis, and developing a new wind energy classification scheme based on the Delphi method, taking in account wind energy, environmental risk and cost factors. These authors emphasised that the potentially richest areas in offshore wind resources are primarily located in the mid- and high-latitudes in far offshore areas, particularly in the regions of the Southern Hemisphere westerlies, the North Atlantic westerlies, and in the Alaska Peninsula. But some regions with high offshore wind resource potential were also identified at lower latitude, namely the Caribbean Sea, the south of the Hawaiian Islands, the Somalian coast and western offshore Peruvian areas. These results are in general aligned with previous investigations like Zheng and Pan (2014) and NREL (2005). A careful analysis of all the latter studies presents a common feature regarding the offshore wind resources in the global coastal areas within the EEZs: the regions in the low mid-latitudes and sub-tropics with larger values of offshore wind potential are clearly associated with the presence of coastal low-level wind jets (CLLJs). Some of these areas correspond to the important Eastern Boundary Current Systems (EBCS), like in California, Humboldt, Iberia, Northwestern Africa, Benguela and West Australia (Ranjha *et al* 2013, Lima *et al* 2018, 2019, Soares *et al* 2019b). Other CLLJ regions showing enhanced wind resource potentials include the Caribbean (Whyte *et al* 2007), the Findlater jet area in both the Somalian-Oman coasts (Ranjha *et al* 2015) and Madagascar (Moore 2013). Finally, there are other conspicuous regions like the Brazilian Belem coast where a large amount of wind resource is highlighted.

The abovementioned EBCS regions are influenced by strong coastal-parallel winds linked to the presence of a high-pressure system over the ocean and the development of a thermal low inland (Winant *et al* 1988). These wind patterns are enhanced by the sharp pressure gradient and by the land-ocean thermal contrast. In general, CLLJ correspond to localized regions of wind maxima within the marine atmospheric boundary layer, usually below the first 500 m, and often show a marked seasonal cycle with maxima wind values during the summer season. Recently, Lima *et al* (2020b) used the newest and highest resolution ERA-5 reanalysis from the European Centre of Medium-Range Weather Forecasts (ECMWF) to revisit the main features of the global systems of CLLJs. This study revealed the benefits of higher horizontal, vertical, and temporal resolution to better represent these coastal wind features. Additionally, many other studies also highlighted the importance of high model grid resolution for onshore

wind energy resource, particularly over complex orography and coastal regions (e.g. Pryor and Barthelmie 2011, Hueging *et al* 2013, Reboita *et al* 2018, Nogueira *et al* 2019). The latter is due to an improved simulation of local circulations associated with land-ocean transitions, such as land/sea breezes and shallow convection, which will also impact the wind patterns in offshore coastal regions examined here. In this sense, the future avenue for offshore wind energy resource assessment requires models with high horizontal and vertical resolution and high temporal sampling.

The present study takes advantage of the recently released ECMWF ERA-5 reanalysis to characterize the global offshore wind resources. ERA-5 is considered the best portrayal of the 3D structure of the atmosphere, featuring significant improvements in spatial and temporal resolution, model parameterization and data assimilation methodologies compared to its predecessors (Hersbach *et al* 2018), thus opening the possibility of characterizing the global offshore wind resource in an unprecedented way. Specifically, our main goals are: (1) characterizing the global offshore wind resource available at a standard turbine height (100 m) within the EEZ coastal areas; (2) depict how relevant are the coastal wind jet regions, within the EEZ, for the wind power potential; (3) demonstrate the usefulness of having vertical model levels information and hourly sampling, and (4) showing the gains connected to converging to 250 m hub heights in offshore wind harvesting in the EEZ regions.

2. Data and methods

2.1. Reanalysis ERA-5

ERA-5 is the fifth-generation global atmospheric reanalysis produced by the ECMWF, extending from 1979 to present, and is being updated within 3 months of real time. Compared to its predecessors, ERA-5 uses a more recent version of the ECMWF Integrated Forecast System model (IFS 41r2), with increased temporal output, horizontal and vertical resolutions (respectively, 1 h, 0.25° and 137 vertical levels extending from the surface to 0.01 hPa), and several improvements to different parameterizations (e.g. convection and microphysics) and to the data assimilation scheme (see Hersbach *et al* (2018) for a detailed description of ERA-5). Recent studies have shown that these improvements result in an overall increased accuracy of ERA-5 compared to its predecessors in representing several climate variables, including rainfall and tropical convection (Beck *et al* 2019, Nogueira 2020), land surface temperature (Johannsen *et al* 2019), downwelling solar radiation at the surface (Urraca *et al* 2018), but also wind patterns of global oceans (Ramon *et al* 2019, Rivas and Stoffelen 2019) and CLLJ (Lima *et al* 2020b).

In the present study, the hourly u - and v -components of wind speed from ERA-5 model levels, for the 1979–2017 period, are used to compute the

wind speed at each model level, which it is then used to interpolate the wind speed at a specific height (the wind speed retrieval methods are described in the next section). Additionally, the near-surface u - and v -components of wind speed at 10 m height are also used.

2.2. Methodology for hub-height wind speed

Three different methods are applied to obtain the wind speed at 100 m height. The first consists in interpolating linearly the wind speed from the ERA-5 model levels output. In the second and third methods, the 10 m wind speed from ERA-5 is used to extrapolate the wind speed to 100 m using two different approximations. One uses the logarithm wind profile extrapolation (Yamada and Mellor 1975), which is defined as $v_z = v_{10} \ln\left(\frac{z}{z_0}\right) / \ln\left(\frac{10}{z_0}\right)$. The other uses the power-law wind profile, which is given by $v_z = v_{10} \left(\frac{z}{10}\right)^\alpha$. Here v_{10} corresponds to the 10 m wind speed and v_z represents the wind speed at a given height z . The α coefficient is a function of the atmospheric stability conditions and it can be determined according to Panofsky and Dutton (1984) as $\alpha = \left[\ln\left(\frac{10}{z_0}\right)\right]^{-1}$. A variable α coefficient is more suitable to use in the power-law wind profile computation, instead use a constant value that assume neutral or near-neutral atmosphere stability conditions, which it is not true in many of the studied regions. The z_0 is the local roughness length, which is computed taking into account the variation of the drag coefficient with the 10 m wind speed (Yelland and Taylor 1996), ranging from 0.036×10^{-3} to 2.24×10^{-3} m.

The wind power density, WPD, is computed as $WPD = \frac{1}{2} \rho v_z^3$, where ρ is the air density (here considered constant and equal to 1.22 kg m^{-3} , according to a standard atmosphere from the International Organization for Standardization). To depict the WPD with a non-constant air density value, the WPD at 100 m height was also computed with a variable air density obtained from ERA-5 model levels.

The WPD is computed at hourly and daily scales considering only wind speed magnitudes between 3 ms^{-1} and 25 ms^{-1} at 100 m height. These cut-off values reflect the typical range in which wind turbines can produce electrical power. A rated regime is not considered here since the analysis is not focused on a specific turbine wind power generation.

The WPD is computed and firstly presented over the full extent of the EEZs (global maps), and secondly for the ocean areas within the EEZs up to 1000 m depths (scatter plots), for the 1979–2017 period. A first analysis is performed for the WPD at 100 m height, followed by an analysis of the sensitivity of estimated 100 m WPD with different wind speed retrieval methods. Finally, the impact of future increases of hub-height on global offshore wind power resources is shown through the relative

differences of the mean WPD available at 100 m and 250 m heights using model levels linear interpolation.

3. Results

3.1. Present climate wind resource

The annual and seasonal averaged 100 m WPD over the global EEZs are depicted in figures 1(a) and 2, respectively. The largest annual WPD values in the northern hemisphere (NH) occur in the mid-latitude storm track regions, ranging between 1200 Wm^{-2} and 1600 Wm^{-2} . As expected, WPD in NH storm-tracks is largest during the boreal winter (DJF, reaching values above 2000 Wm^{-2}) and lowest during boreal summer (JJA, falling below 800 Wm^{-2} throughout most of the EEZs). Annual averaged WPD values above 800 Wm^{-2} are also found at higher latitudes, particularly in the Greenland Sea and Norwegian Sea (figure 1(a)). These features are related to the Icelandic low and to frequent cyclonic activity, particularly the occurrence of polar lows (Liu *et al* 2016, Przybylak 2016), thus also peaking during DJF and displaying minimum values during JJA (figure 2). The remaining Arctic offshore regions are generally characterized by moderate WPD values, between 400 Wm^{-2} and 800 Wm^{-2} (figure 1(a)). These are associated with the large-scale geostrophic flow where the most frequent pattern is the polar easterlies (Przybylak 2016, Liu *et al* 2016), which are characterized by a less pronounced seasonal cycle (figure 2). The seasonal maximum WPD value in the NH is localized south of Greenland, surpassing 1600 Wm^{-2} , which is a signature of the tip Greenland tip jets (Doyle and Shapiro 1999, Moore and Renfrew 2005). These local wind accelerations, stronger during winter, are due to the interaction between cyclonic activity and the Greenland's orography and, consequently, weakest during summer (figure 2). Another confined region of relatively high annual averaged WPD ($>800 \text{ Wm}^{-2}$) emerges in the Mediterranean, corresponding to the Mistral wind, blowing from southeastern France. Figure S1 (available online at stacks.iop.org/ERL/15/1040a2/mmedia) shows the relative differences of the annual mean WPD at 100 m height computed with a variable air density obtained from ERA-5 model levels and using the constant value. Overall, the differences between computing the wind power density with a variable air density and with constant values are small, however existent. At the tropical regions, the relative differences can reach -5% , and at high latitude between 7.5% and 10% . At the sub-tropics and mid-latitude regions, the relative differences are close to zero.

In the southern hemisphere (SH), at latitudes poleward of about 40°S , the SH westerlies cause values of annual averaged WPD above 400 Wm^{-2} (often higher) on the southern coastal regions of South America, Oceania and Africa, peaking offshore of

Cape Horn (South America), South East Cape (Australia) and south of New Zealand with values over 1600 Wm^{-2} (figure 1(a)). In general, the magnitude of WPD associated with the SH westerlies has low seasonal variation (figure 2). The main exceptions are the maxima south of New Zealand and Cape Horn which are amplified during boreal spring (MAM) and boreal autumn (SON), although their magnitude remains very large throughout the entire year. The local maximum offshore of South East Cape does not show relevant seasonal variations.

WPD is generally below 400 Wm^{-2} in the tropics and sub-tropics. However, there are some exceptions, corresponding to confined regions where annual averaged values raise above 800 Wm^{-2} . One of those exceptions is the China Seas region, where annual averaged WPD is $>400 \text{ Wm}^{-2}$ peaking in the Taiwan Strait with values $>1200 \text{ Wm}^{-2}$, associated with subtropical western Pacific winter monsoon and its interaction with local orography (Sampe and Xie 2007), which also explains the larger (lower) values in DJF (JJA). Most of the other exceptions correspond to regions where CLLJs occur, namely: California, Benguela, Canaries, Caribbean, Somalia-Oman and, with in less extent the Western Australia and Peru-Chile CLLJs. Figure 2 shows that the WPD seasonal patterns over these regions reflect quite clearly the seasonal cycle of CLLJs. In California, the higher persistency of the CLLJ in JJA (Lima *et al* 2018, 2020b), leads to the maxima values of WPD, between 1200 Wm^{-2} and 1600 Wm^{-2} , and in a localized area up to 1900 Wm^{-2} . In the EBCS of Canaries, the largest WPD values occur in the southern areas offshore of Senegal during MAM, and in the northern areas offshore of Morocco (Cape Ghir) during JJA, following closely the seasonal cycle of the CLLJ (Lima *et al* 2018, 2020b, Soares *et al* 2019b). In the western coast of Iberian Peninsula, the maximum WPD is found downwind of the main capes, where the available wind resource does not exceed 900 Wm^{-2} . Another remarkable case concerns the Oman/Somalian jet region, which displays one of the largest seasonal cycles worldwide of energy density, directly linked to the very persistent and intense CLLJ in JJA, which forces values close to 2500 Wm^{-2} . This large summer peak is associated to the Findlater Jet (Findlater 1969), a cross-equatorial flow resulting from the interaction between the high-pressure system over the Indian Ocean and the thermal-low over Asia. In turn, the interaction between the Findlater Jet and the South Asia summer monsoon results in a south-westerly coastal parallel wind along the southeast Arabian Peninsula coast, developing the Oman CLLJ. In the other seasons, the WPD does not surpass the 400 Wm^{-2} , except for a small region offshore of Somalia during SON. The south Caribbean Sea also presents high values of WPD, up to 2000 Wm^{-2} , in boreal winter, and between 700 Wm^{-2} and 1200 Wm^{-2} in the other seasons. This

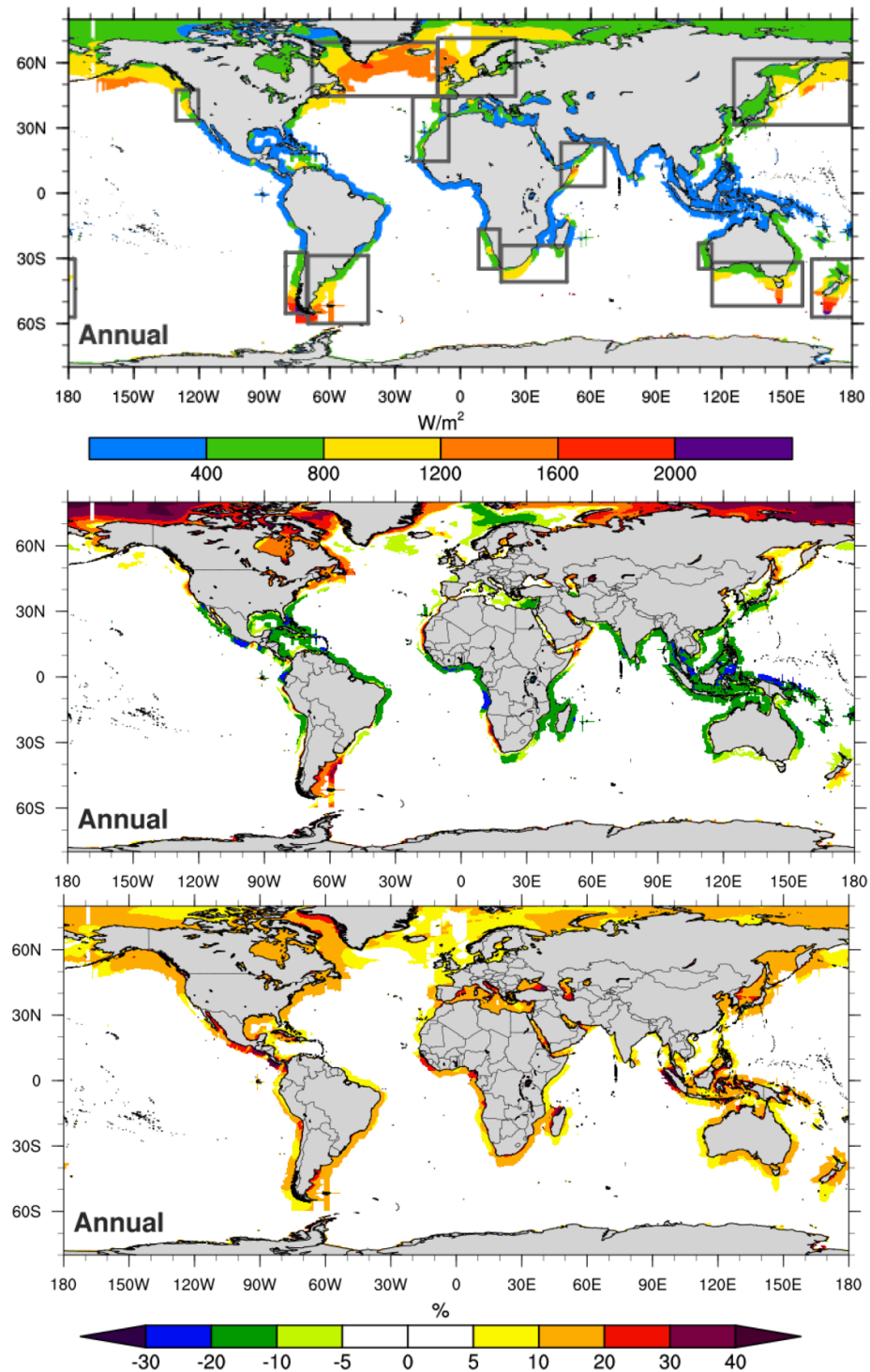


Figure 1. (a) Annual averaged mean power density (Wm^{-2}) at 100 m from ERA-5 (1979–2018) based on model levels linear interpolation, within the economic exclusive zones. Relative differences of the annual mean power density (Wm^{-2}) at 100 m height between (b) model levels linear interpolation and the logarithm approach and (c) computed at an hourly and daily scale. The boxes delimit regions where a special analysis is performed.

region is characterised by an intensification of the trade winds due to the zonal sea surface temperature gradients between the warm eastern equatorial Pacific and cool equatorial Atlantic (Whyte *et al* 2007). Along the Chilean-Peruvian coast, the higher availability of wind resource occurs during austral summer season, where the maximum WPD is up to 1200 Wm^{-2} . During the other seasons the available resource is

below 800 Wm^{-2} . The Benguela CLLJ is stronger and more persistent in austral spring and summer (Lima *et al* 2018, 2019, 2020b), and accordingly, the WPD amounts up to 1600 Wm^{-2} . The West Australian coast, the available resource has high variability, with a maximum of WPD around 1000 Wm^{-2} during austral summer and decreasing to 400 Wm^{-2} in winter season.

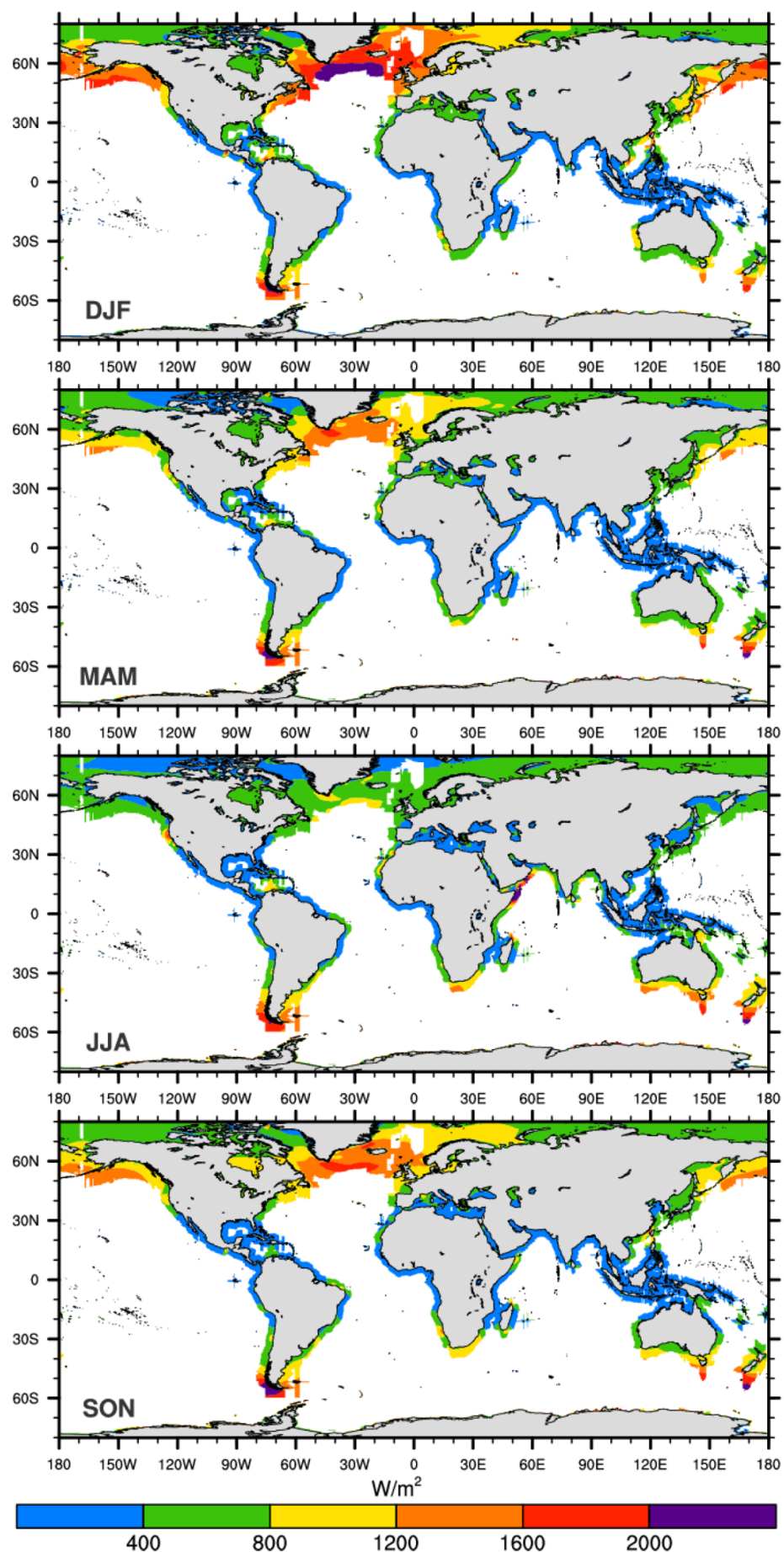


Figure 2. Seasonal averaged mean power density (W m^{-2}) at 100 m from ERA-5 (1979–2018) based on model levels linear interpolation, within the economic exclusive zones. DJF—December, January and February, MAM—March, April and May, JJA—June, July and August, SON—September, October and November.

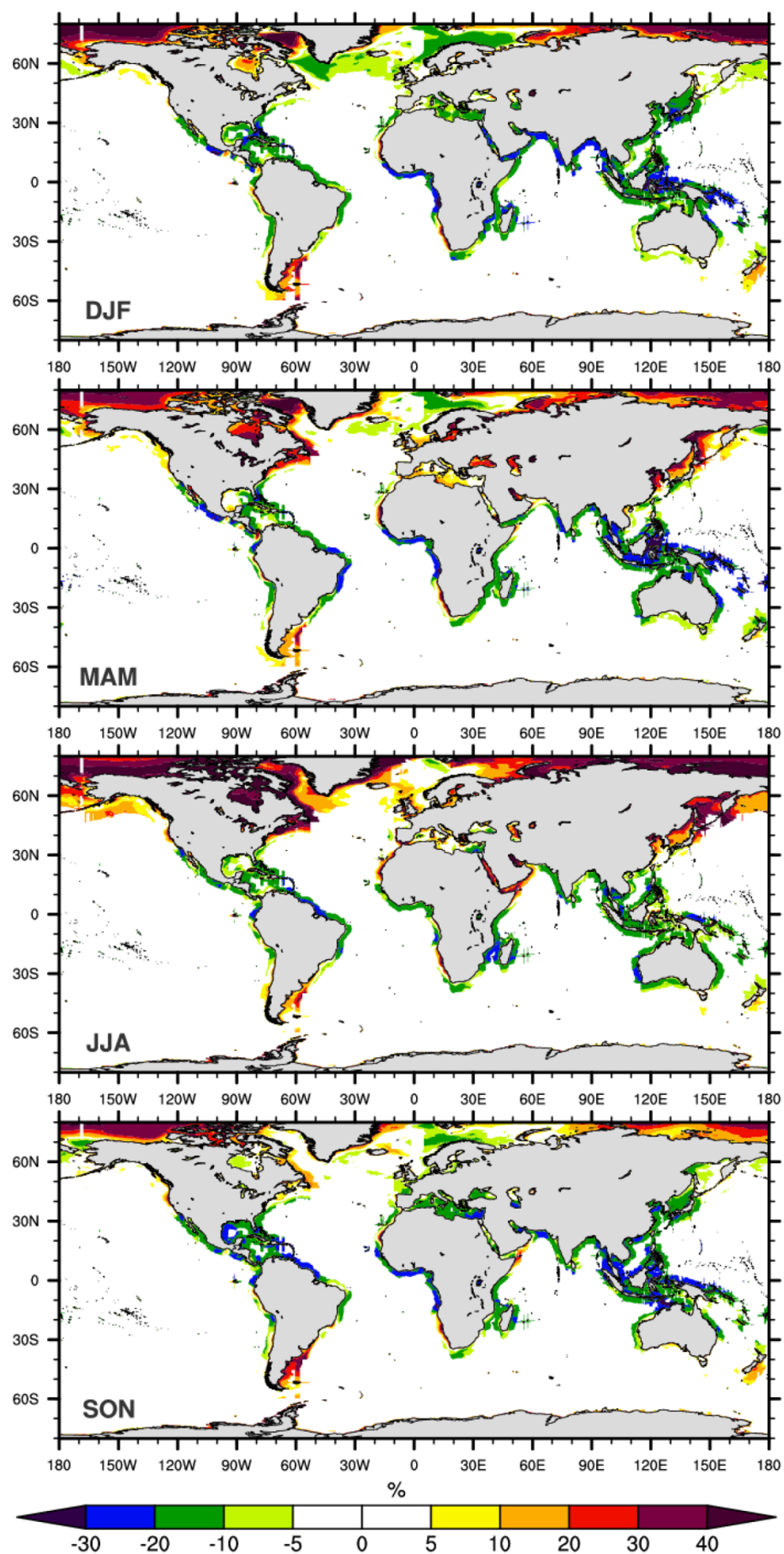


Figure 3. Relative differences of the seasonal mean power density (%) at 100 m height between model levels linear interpolation and the logarithm approach, within the economic exclusive zones. DJF—December, January and February, MAM—March, April and May, JJA—June, July and August, SON—September, October and November.

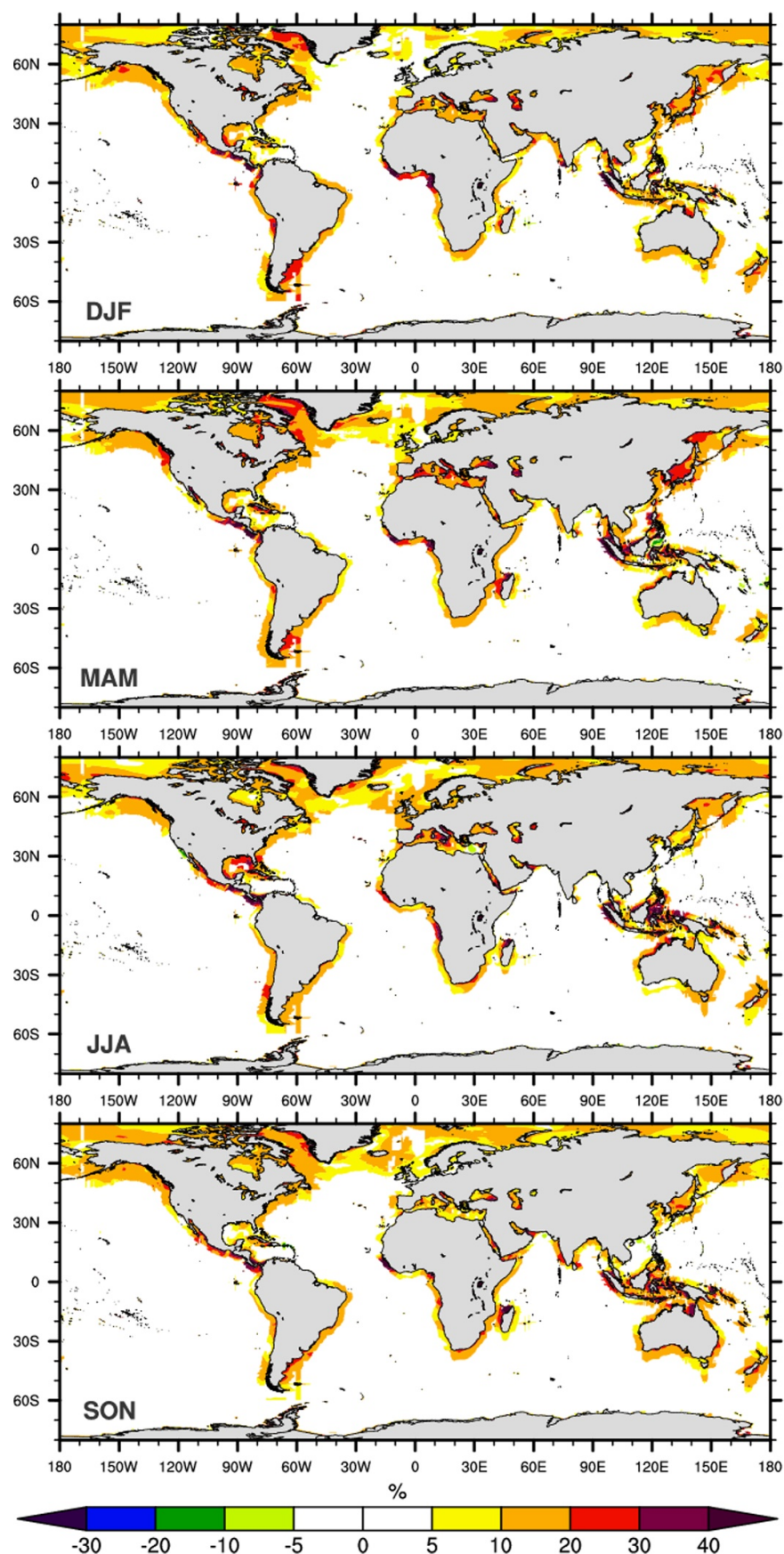
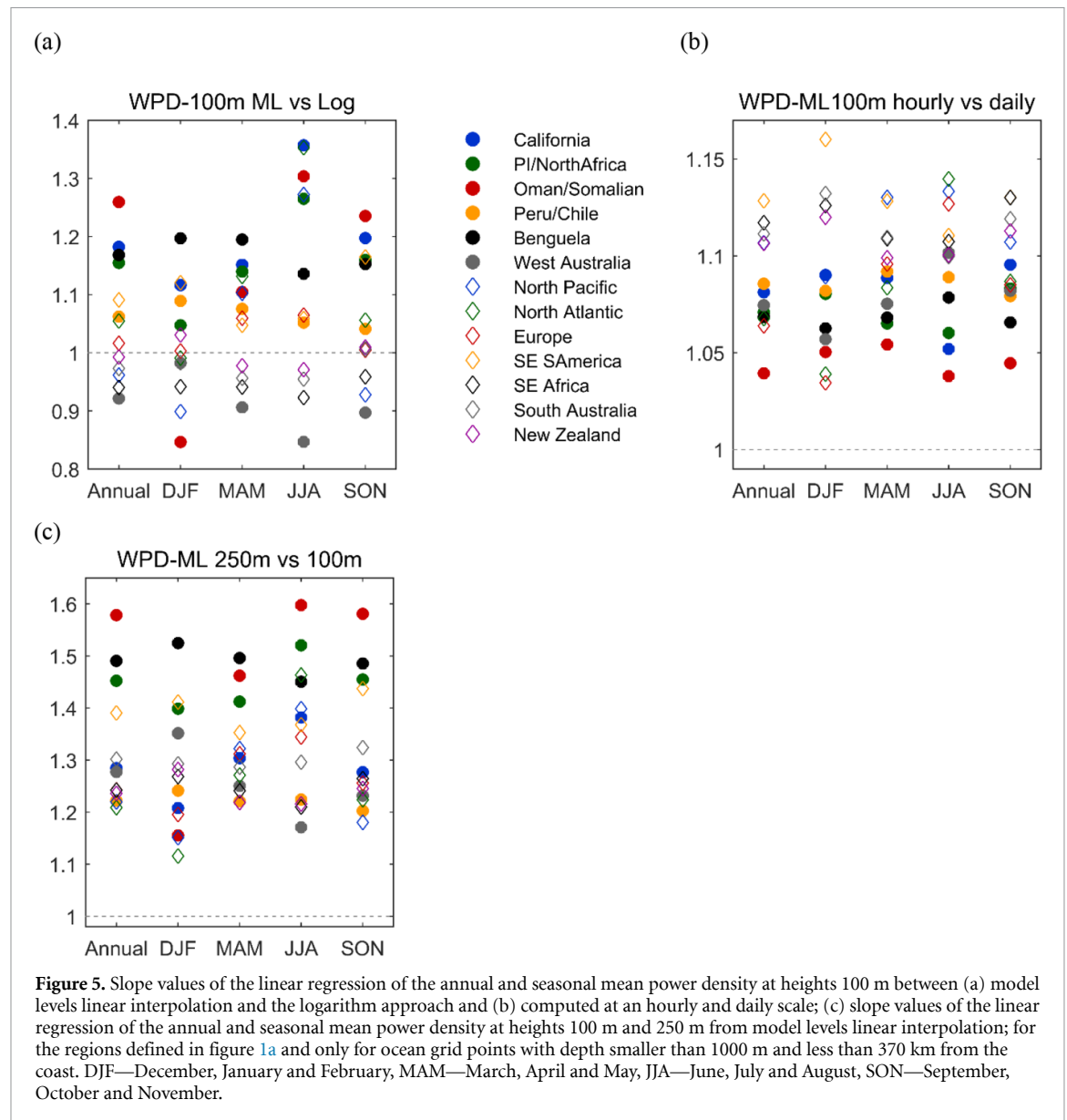


Figure 4. Relative differences of the seasonal mean power density (%) at 100 m height between computed at an hourly and daily scale, within the Economic Exclusive Zones. DJF—December, January and February, MAM—March, April and May, JJA—June, July and August, SON—September, October and November.

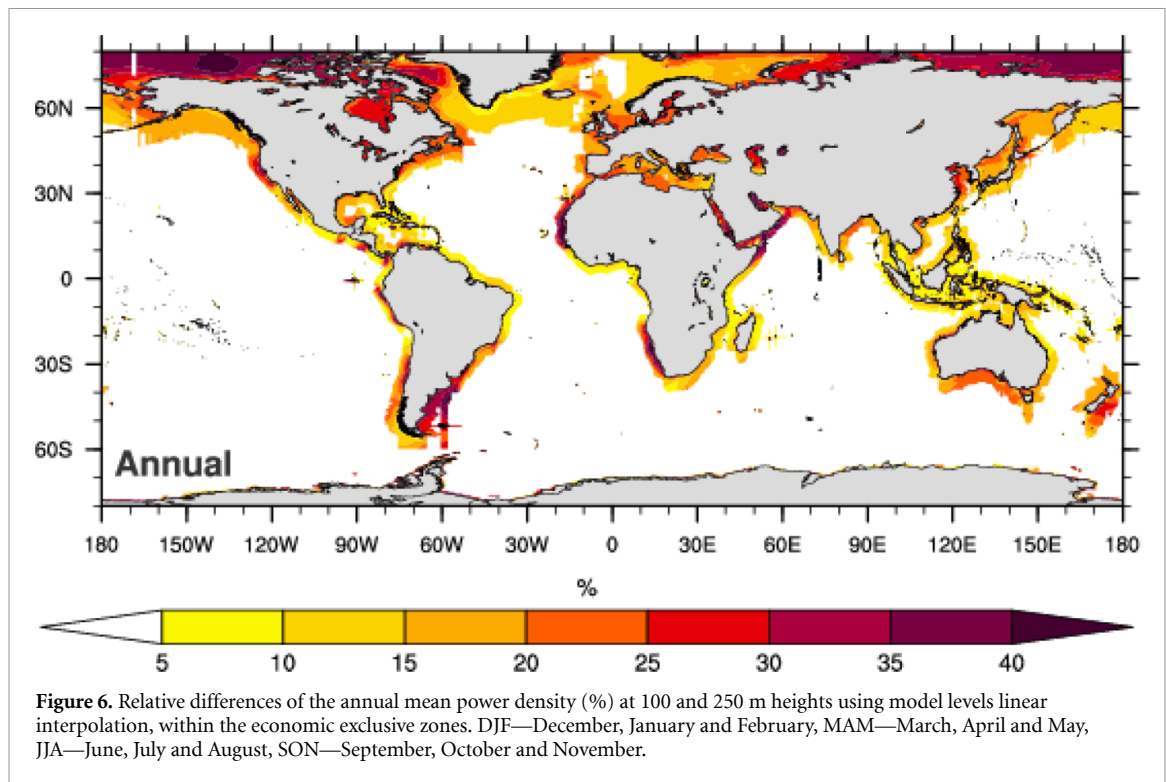


3.2. Sensitivity of estimated wind power density to wind speed retrieval method

The sensitivity of WPD to wind speed retrieval method is assessed for the EEZ regions. We computed the differences of WPD using linear model levels interpolation (assumed here to be the baseline) against the logarithm extrapolation from 10 m to 100 m (figures 1(b) and 3). The impact of the logarithm interpolation for annual averaged WPD is small over the large wind power resource regions dominated by large-scale flow, namely the NH hemisphere storm track and SH westerlies (figure 1(b)). But the logarithm approximation underestimates WPD over most of the Arctic by more than 20%, and often more than 30%. In the tropics and sub-tropics, the logarithm approach overestimates the WPD in extended coastal areas by more than 10% (and up to ~30%). Additionally, the logarithm approach underestimates WPD by more than 20% (reaching more than 30% in localized areas) in the regions of the CLLJs,

particularly for the cases of the Canaries, Benguela, Oman and California. A comparison between figures 2 and 3 shows that this underestimation in the CLLJ regions is roughly proportional to the CLLJ magnitude. In other words, the logarithm interpolation fails to reproduce the large intensification of the wind speed at higher altitudes in the CLLJ regions, as previously suggested by (Soares *et al* 2019a).

Interestingly, figure 1(b) shows that the logarithm approximation also underestimates WPD over mid-latitude eastern coasts of relatively high resource, namely in Canada and northeastern U.S., Siberia, Argentina and New Zealand (but not in Africa). The eastern coast underestimation is even larger when the seasonal cycle is considered (figure 3). For example, the northeastern Asian EEZ shows widespread underestimation of more than 30% during MAM and JJA when the logarithm approximation is considered. Similarly, underestimation is also found for the northeastern U.S. and eastern Canada coastal



regions during JJA. Another noteworthy feature of the seasonal cycles of the differences between logarithmic and model level interpolations is the change of the signal between DJF and JJA in the Labrador Sea. In contrast, the logarithm approach does capture well some local features such as the Greenland jets and the Mistral wind. A detailed explanation of all these features would require an extensive analysis of the circulation and boundary layer structure over each of these regions, which is beyond the scope of the present manuscript.

Using the power-law extrapolation for wind speed resulted in similar difference patterns to the logarithm extrapolation, both at annual (figure S2) and seasonal (figure S3) scales. In other words, both the logarithm and power-law extrapolations of 10 m wind speed to 100 m resulted in poor estimates of the WPD compared to interpolation from ERA-5 model level information due to the difficulty in representing in an appropriate manner the atmospheric stability conditions.

A regional view of the differences between the logarithm extrapolation and model-level interpolation approach for WPD estimation is given for a chosen set of regions identified in figure 1(a), and constrained for EEZ coastal points where depth is up to 1000 m. The differences are illustrated in figure 5(a) by the slopes of the linear regressions of annual and seasonal mean 100 m energy densities between the two approximations. At the annual scale, the coastal and shallow areas where the logarithm most fails are again the ones where CLLJ are persistent, such as Oman/Somalia, California, Benguela

and PI North Africa and Peru/Chile (with values from 1.27 to 1.08). The seasonal cycles are not large in most of the considered regions, as aforementioned. The exceptions are the CLLJ regions of Oman/Somalia, California and PI North Africa, which are characterized by enhanced flow during boreal summer, and also during autumn in the case of the Oman/Somalia CLLJ.

Subsequently, the impact of the model output temporal sampling on WPD estimation is assessed. The underlying motivation for this analysis is the non-linearity of turbine power curves due to cut-off associated with high and low wind thresholds (see section 2) and the marked wind diurnal cycle in some coastal regions. Figures 1(c) and 4 depict the differences of WPD when considering hourly and daily wind speed values, at annual and seasonal scales, respectively. With no exceptions, it is evident that daily wind speed values lead to an underestimation of the wind power resource worldwide in coastal regions, for both annual and seasonal averages, ranging between 5% and 30% over most of the global EEZs. However, a comparison between figures 1(a) and (c), and between figures 2 and 4, shows that the largest overestimations due to coarser temporal sampling occur over regions of relatively low annual averaged wind power resource, while over the regions of large WPD the errors are smaller than 20% (and often between 5% and 10%).

Figure 5(b) shows the impact of the temporal sampling on WPD aggregated for the set of regions identified in figure 1(a) constrained to points within

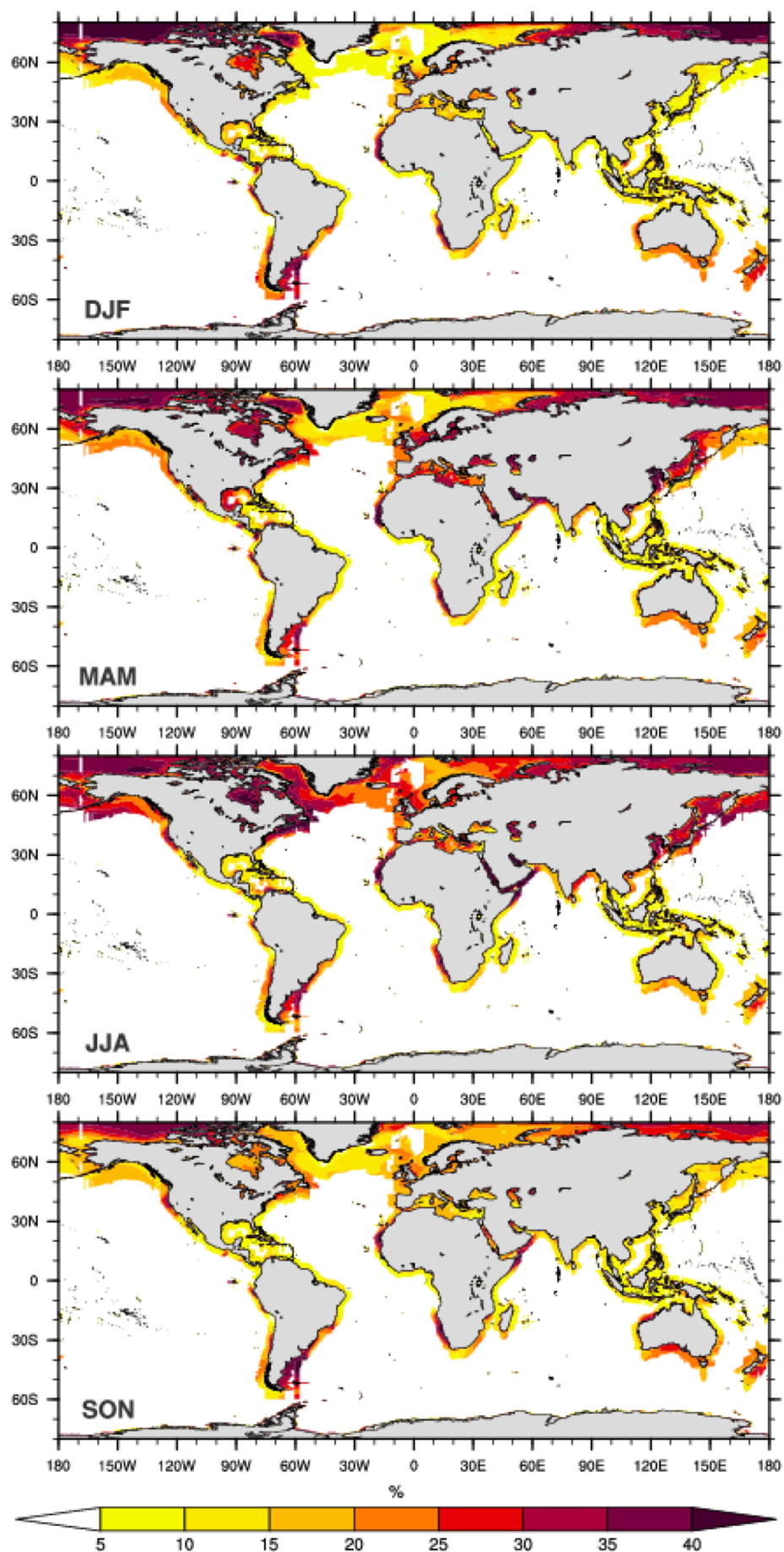


Figure 7. Relative differences of the seasonal mean power density (%) at 100 and 250 m heights using model levels linear interpolation, within the economic exclusive zones. DJF—December, January and February, MAM—March, April and May, JJA—June, July and August, SON—September, October and November.

EEZs and water depths below 1000 m. The slope values of the linear regression of the annual and seasonal mean 100 m WPD between hourly and daily values reveal WPD gains for all regions when hourly wind speed output is considered w.r.t. to daily output. At the annual scale, the vast majority of the EEZ areas show between 10% and 20% more WPD when hourly output is considered. Further, the gains are even larger mostly in some tropical and sub-tropical regions such as in offshore areas of West Africa, in Indonesia and southern Mexico pacific coast, where the hourly sampling leads to more WPD (between 10% and 20%). In general, this is also true at seasonal scale, but the gains can reach values above 30% in some localized coastal areas (figure 4). Looking for the regional aggregated values of the gains associated to the hourly output w.r.t. daily, for the EEZ coastal points where depth is up to 1000 m (regions delimited in figure 1(a)) it can be observed that those are relatively small, ranging from 1.02 and 1.16, *i.e.* to 2%–16% (figure 5(c)).

3.3. Impact of increasing hub-height on offshore wind power density

The annual averaged change to WPD due to increasing hub height to 250 m w.r.t. 100 m is shown in figure 6. The results show largest gains (>30%) in offshore regions with high wind resource at 100 m: the Arctic Pacific sector, eastern coast of Argentina, and in different CLLJ regions (e.g. California, Humboldt, Benguela, Canaries and Oman). The latter are expected since the CLLJ wind speed maxima occur at altitudes between 300 m and 500 m. Therefore, the height gains are more prominent and spatially confined along the EBCS regions and can reach e.g. 50% along the Northwestern Africa and Benguela regions, at the annual scale. In agreement with the average WPD seasonal cycles, the seasonal WPD gains due to increased hub-height display larger magnitude and display higher spatial variability than the annual averages (figure 7). Milder gains, above 5% and 10%, are rather common throughout most of the global coastal areas.

In figure 5(c), the relationship between the WPD at 250 m and 100 m is shown for each region defined in figure 1(a) limited to the EEZs and to 1000 m seafloor depths. In agreement with the previous results, the annual and seasonal slope values are above 1 in all regions considered; the latter results mean that at 250 m is available much more WPD. The higher values are especially significant in some of the CLLJ regions, such as Oman, Benguela, PI/North Africa, California, where annual slope values are well above one (in the range of 1.35 and 1.62). Responding to the CLLJs seasonal cycle, when the coastal jet is more intense larger slope values can be identified, like in boreal winter and spring in Benguela (~ 1.55), in boreal autumn in Oman/-Somalia (~ 1.7), in boreal summer in California and

PI/North Africa (~ 1.5 and 1.62). Showing smaller but important slope values are also non-jet areas like Peru/Chile and South Australia, where slopes amount to around 1.3.

4. Conclusions

Offshore wind power has been developing rapidly over the last years, but its full potential is still far from being achieved. The present study tackles two important topics for future planning and development of offshore wind power resources: accurate quantification of available wind resources over the global EEZs, including its spatial and temporal variability; and quantification of potential gains from technological developments allowing for higher hub heights around 250 m (compared to the current typical ~ 100 m values).

We present a global characterization of WPD resource based on the ERA-5 reanalysis. We identified the main regions with large offshore wind power potential, relate them to the underlying atmospheric circulation features and characterized its seasonal variability. The global WPD quantitative estimates should represent a new (updated) benchmark, given the significant improvements in the ERA-5 reanalysis compared to its predecessors (particularly high spatial and temporal resolution, improved data assimilation scheme and increased amount of assimilated observations). One argument to support this claim is that many of the EEZs with large wind power potential are characterized by regional-scale circulation features, such as CLLJs, or correspond to coastal regions where larger-scale circulations (such as the large-scale easterlies and westerlies, or cyclones and anti-cyclones) interact with the complex land-sea contrasts and topography. The impact of increased model resolution on the description of these complex interactions has been well established over the past decades. Additionally, the ERA-5 WPD was consistent with robust results presented by previous studies (NREL 2005, Zheng and Pan 2014, Sasaki 2017, Zheng *et al* 2018). Namely, our results showed WPD higher than 800 Wm^{-2} in the mid- and high-latitudes, and higher than 400 Wm^{-2} along the EBCS. This is in agreement with the wind energy classification presented in the abovementioned studies, where these regions belong to class 7. Another argument to support our claim is our demonstration of the large impact of WPD estimation directly from model levels in a dataset with high vertical resolution, compared to the often-used log and power-law extrapolation of the near-surface wind speed. We showed that the model level estimation can give rise to differences of 50% in estimated wind power resource compared to the extrapolation methods. The results also suggest that these differences will be hard to correct by post-processing techniques given their, physical-process dependent, large spatial and temporal heterogeneities. Furthermore,

we also demonstrated the positive impact of considering hourly temporal resolution when compared to the, also often-used, daily resolution. An accurate representation of the sub-daily variability can have large impact due to turbine wind cut-off thresholds. Indeed, our results showed widespread underestimation of daily based WPD estimates by more than 10%, and up to around 30%.

Taking advantage of the very high vertical discretization of ERA-5 within the maritime planetary boundary layer, we assessed the impact of future technological developments for higher hub-heights. The results showed large positive impact for global offshore wind farming from increasing hub-heights to 250 m, with widespread gains (and no losses) across all EEZs and all seasons, ranging between 5% and 50%. Moreover, these gains are >10% for all regions with large wind power potential and >40% over the regions with CLLJs and the Arctic Pacific sector.

Finally, all the depicted differences and gains are especially relevant in the context of the several wind power climate change assessment studies recently published (Soares et al 2017, 2019a, Keivanpour et al 2019, Lima et al 2020a). In fact, these studies point out to future wind power changes of roughly the same magnitude as the ones here studied.

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Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

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